Day 4 Task: CyberChef Lab

Part A — Decode the hidden messages

#	Decoded message	Operation(s) used
A1	Welcome to Tekloomy. Encoding is not encryption.	From Base64
A2	Hash everything you download.	From Hex
A3	Zero to Defender	From ROT13
A4	Backups beat ransomware	From Base64, then From Hex

Part B — Hashing

Input (type exactly)	First 8 chars of SHA-256
tekloomy	654b8d2e
Tekloomy	53c9cd59
tekloomy (with a space at the end)	f6636caf

B4. One capital letter changed. How much of the hash changed? What does this tell you about why hashes are used to verify downloads?

The entire hash changed completely. This shows that even a tiny change in a file produces a radically different hash, making it extremely easy to detect tampering or corruption.

Part C — Inspect a real certificate

Field	What you found
Issued to (Common Name / domain)	*.google.com
Issued by (Certificate Authority)	CN = WE2, O = Google Trust Services ,C = US
Valid until (expiry date)	12/31/26, 12:37:41 AM GMT+5:30
Public key algorithm (e.g. RSA, ECC) and size	Elliptic Curve Public Key

C5. Now visit https://expired.badssl.com. What warning do you see, and what does it actually mean? Should you click 'Proceed'?

Warning: "Your connection is not private" (NET::ERR_CERT_DATE_INVALID). Meaning: The website's certificate has expired, so the browser cannot securely verify its identity. You should NOT proceed as the connection is no longer guaranteed to be secure.

Part D — Encrypt for real

D1. Paste the encrypted output (hex):

96 44 0f 5c 86 7b 0e 90 14 9a 7d 94 82 21 20 5a

D2. Change one character of the key and try to decrypt. What happens? Why does this matter for ransomware?

The decryption completely fails and produces an error or gibberish. This matters for ransomware because if you do not have the exact, perfect decryption key, it is mathematically impossible to recover your locked files.